

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Eastern Bluebird"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

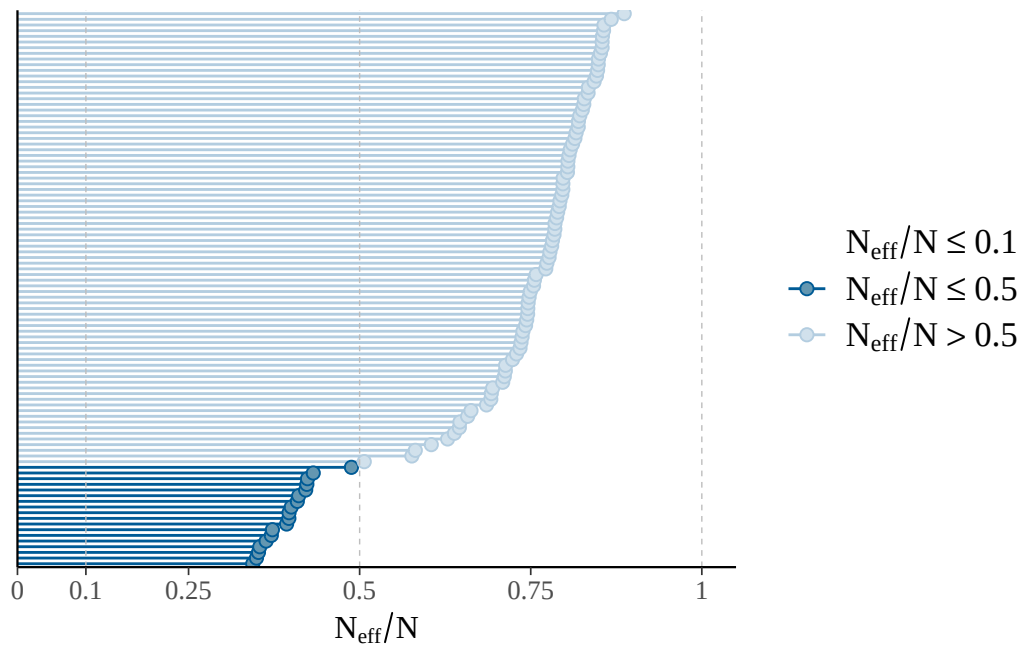
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3438	0.7890	0.9671	1.0038	1.1681	2.0916

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4539	0.6901	0.7566	0.7382	0.8119	0.8866

```
[1] "effective sample size in bulk and tail fine"
```

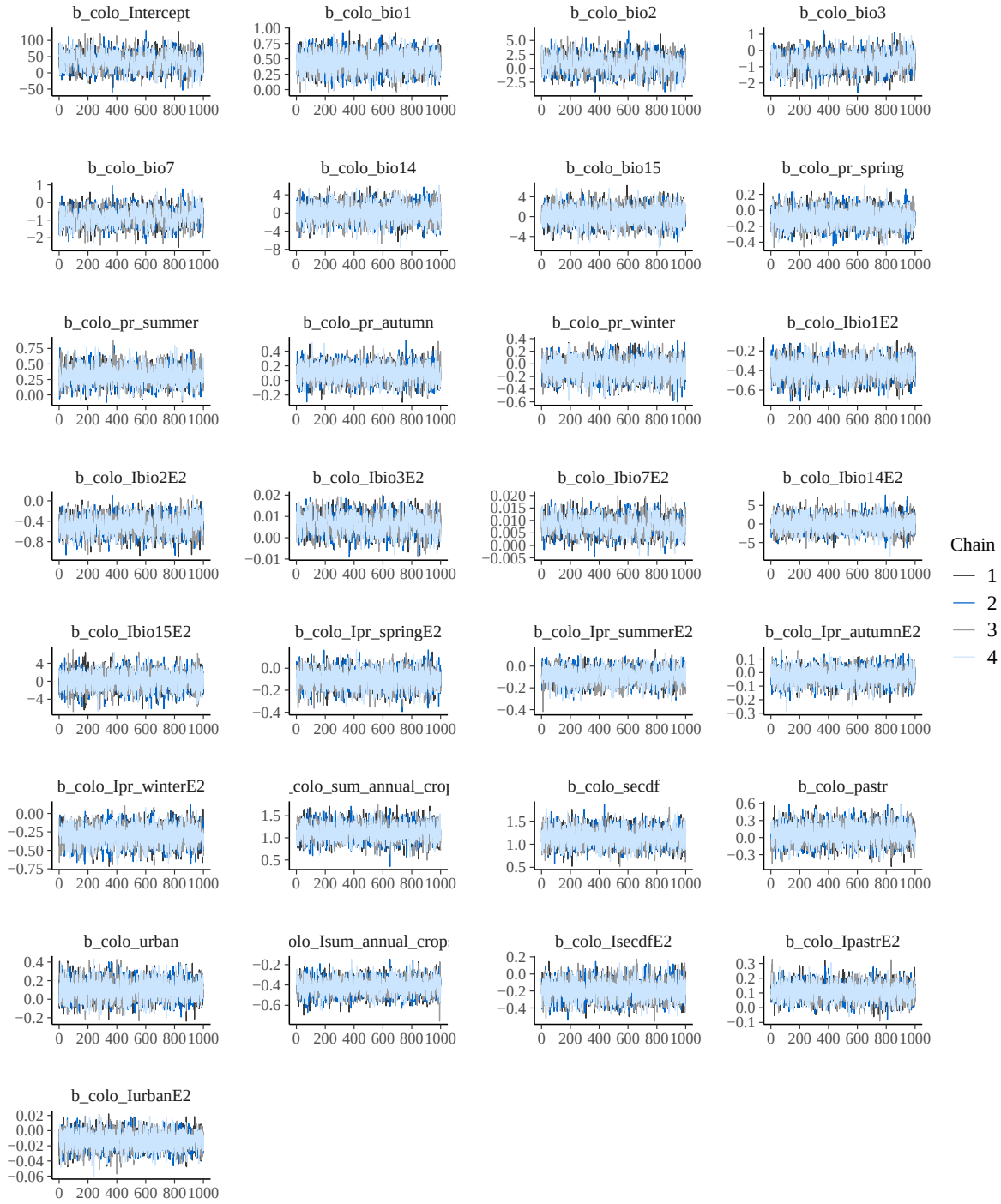
Do chains mix well:

No divergences to plot.

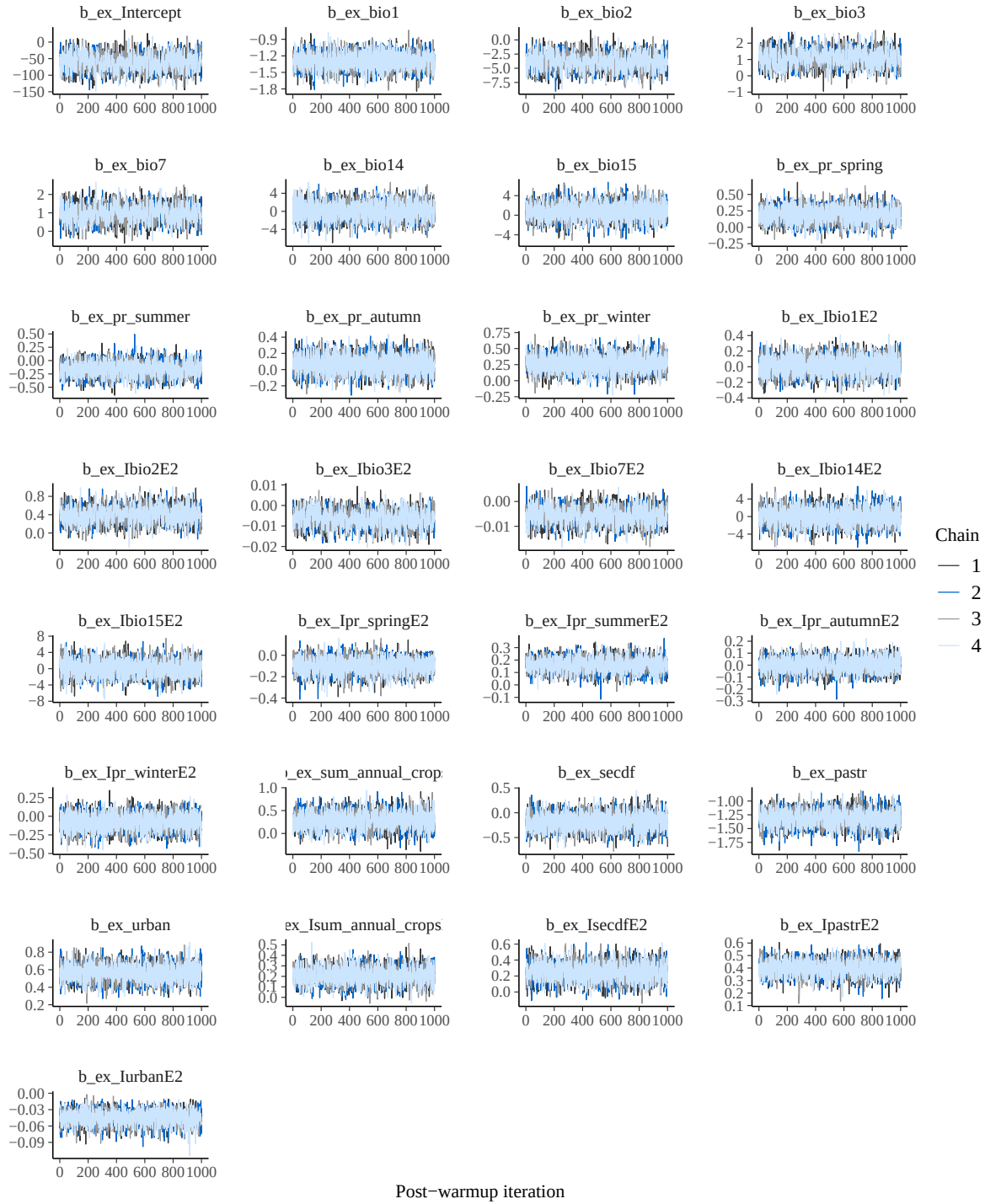
No divergences to plot.



No divergences to plot.



Post-warmup iteration



Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9996	1.0001	1.0003	1.0007	1.0044

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"          "b_occ_Ibio3_3yrsE2"
[3] "b_occ_Ipr_summer_3yrsE2" "b_occ_Ipr_autumn_3yrsE2"
```

```
[1] "b_colo_bio1"              "b_colo_pr_summer"
[3] "b_colo_Ibio1E2"           "b_colo_Ibio2E2"
[5] "b_colo_Ibio7E2"           "b_colo_Ipr_winterE2"
[7] "b_colo_sum_annual_crops"  "b_colo_secdf"
[9] "b_colo_Isum_annual_cropsE2" "b_colo_IpastrE2"
```

```
[1] "b_ex_Intercept"          "b_ex_bio1"
[3] "b_ex_bio2"              "b_ex_bio3"
[5] "b_ex_bio7"              "b_ex_pr_winter"
[7] "b_ex_Ibio2E2"           "b_ex_Ipr_summerE2"
[9] "b_ex_pastr"             "b_ex_urban"
[11] "b_ex_Isum_annual_cropsE2" "b_ex_IsecdfE2"
[13] "b_ex_IpastrE2"          "b_ex_IurbanE2"
```

